

MA 102 Mathematics II
Ordinary Differential Equations
Lecture 11
(April 1, 2024)

Second-order ODEs and singularities

The general homogeneous second-order linear equation is of the form

$$y'' + P(x)y' + Q(x)y = 0. \quad (1)$$

We already know that equation (1) can be very easily solved for constant P and Q , and also for some specific $P(x)$ and $Q(x)$. For other kinds of $P(x)$ and $Q(x)$, power series solution is (probably) the only analytical procedure.

The main fact about (1) is that its solutions near a point x_0 depends on the behaviour of its coefficient functions $P(x)$ and $Q(x)$ near this point. We restrict ourselves to the case in which $P(x)$ and $Q(x)$ are well-behaved in the sense of being analytic at x_0 , which means that each has a power series expansion valid in some neighbourhood of this point.

In this case x_0 is called an ordinary point of equation (1) and it turns out that every solution of the equation is also analytic at this point. Any point that is not an ordinary point of (1) is called a singular point.

In other words,

x_0 is called an **ordinary point** of a function if the function has a power series expansion in some neighbourhood of x_0 . It is called a **singular point** if the function does not have a power series expansion in any neighbourhood of x_0 . Accordingly we term x_0 to be an ordinary or singular point of the differential equation in which those functions appear as the coefficient functions.

Let us state a theorem which will help us in knowing about the solution of a second order equation:

Let x_0 be an ordinary point of the ODE

$$y'' + P(x)y' + Q(x)y = 0, \quad (2)$$

and let a_0 and a_1 be arbitrary constants. Then there exists a unique function $y(x)$ that is analytic at x_0 , which is a solution of equation (2) in a certain neighbourhood of this point, and satisfies initial conditions $y(x_0) = a_0$ and $y'(x_0) = a_1$.

Furthermore, if the power series expansions of $P(x)$ and $Q(x)$ are valid on an interval $|x - x_0| < R$ ($R > 0$), then the power series expansion of this solution is also valid on the same interval.





Classification of singular points:

Consider x_0 to be a singular point of the equation

$$y'' + P(x)y' + Q(x)y = 0. \quad (3)$$

Many differential equations have singular points and the choice of physically appropriate solutions is often determined by their behaviour near these points. It is interesting to note that although we want to avoid the singular points of a differential equation, it is precisely these points that usually demand particular attention.



Example:

If we consider

$$y'' + \frac{2}{x}y' - \frac{2}{x^2}y = 0,$$

then the origin is clearly a singular point.

$y_1 = x$ and $y_2 = x^{-2}$ are the linearly independent solutions for $x > 0$, so $y = c_1x + c_2x^{-2}$ is the general solution on this interval. Note that the differential equation is in Cauchy-Euler form. But if we want the solution to be bounded near the origin, this can be obtained by setting $c_2 = 0$.

In most of the applications, the singular points are rather “weak” in the sense that the coefficient functions are only mildly non-analytic and simple modifications will yield satisfying solutions.

Definition:

A singular point x_0 of equation (3) is called **regular** if the functions $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic, and **irregular** otherwise.

In other words, the singularity in $P(x)$ cannot be worse than $1/(x - x_0)$, and that in $Q(x)$ cannot be worse than $1/(x - x_0)^2$.



Example:

Consider

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0.$$

Here $P(x) = \frac{-2x}{1-x^2}$, $Q(x) = \frac{n(n+1)}{1-x^2}$ and $x = \pm 1$ are regular singular points.

The recursion relation can be obtained as

$$a_{k+2} = -\frac{(n-k)(n+k+1)}{(k+1)(k+2)}a_k,$$

by using the series solution $y = \sum_{k=0}^{\infty} a_k x^k$. (We will know later how this recursion relation is obtained.)

In most of the second-order equations, we encounter only what are known as **two-term recursion formulas** for the coefficients of the unknown series solutions.

Usually it is fairly easy to determine the general form of the resulting series and to obtain precise information about their radii of convergence.



In most cases, the best we can do is to find the radii of convergence of the series expansions of $P(x)$ and $Q(x)$ and conclude that the radius for the series solution must be at least as large as the smaller of these numbers.

Thus, we can conclude from

$$\frac{1}{1-x^2} = 1 + x^2 + x^4 + \cdots,$$

that $R = 1$ for both $P(x)$ and $Q(x)$ ($P(x)$ is simply the above expression multiplied by x – ignore the sign and the constant). Therefore, without further calculation, we know that any solution of the form $y = \sum a_k x^k$ of this equation must be valid at least on the interval $|x| < 1$.

Definition (Frobenius Series)

A “quasi power series” of the form

$$y = x^m \sum_{k=0}^{\infty} a_k x^k = x^m (a_0 + a_1 x + a_2 x^2 + \cdots) \quad (4)$$

is called a **Frobenius series** and the procedure for finding solutions of ODEs by using this type of power series is called the **method of Frobenius**.

The exponent m can be integer, fraction, positive or negative. Frobenius series is used when we want to solve some differential equations which have regular singular point(s).

Frobenius method will always work provided the point of expansion is no worse than a regular singular point. In (4) above, the coefficients as well as the exponent m are undetermined.

The values of exponent m are determined from an equation known as the **indicial equation**. The coefficients a_k are determined by equating the coefficients of various powers of x . That is, we need to find a recursion relation among the coefficients.



In general, we may assume that the singular point x_0 is located at the origin.

The general form of a function f analytic at $x = 0$ is $f(x) = a_0 + a_1x + a_2x^2 + \dots$

The origin will certainly be a singular point of

$$y'' + P(x)y' + Q(x)y = 0$$

if P and Q have the forms

$$P(x) = \dots + \frac{b_{-2}}{x^2} + \frac{b_{-1}}{x} + b_0 + b_1x + b_2x^2 + \dots,$$

$$Q(x) = \dots + \frac{c_{-2}}{x^2} + \frac{c_{-1}}{x} + c_0 + c_1x + c_2x^2 + \dots,$$

and at least one of the coefficients with negative subscript is nonzero.

If x_0 is a regular singular point, a Frobenius series can be justified as the solutions.

Two linearly independent solutions are possible if, in above expressions

- $P(x)$ does not contain more than the first term to the left of b_0 .
- $Q(x)$ does not contain more than the first two terms to the left of c_0 .

It can be equivalently stated as

$xP(x)$ and $x^2Q(x)$ must be analytic at the origin, and according to the definition, this is precisely what is meant by the singular point $x = 0$ is regular.

Let us consider

$$x^2y'' + pxy' + qy = 0, \quad (5)$$

where p and q are constants.

It can be written as

$$y'' + \frac{p}{x}y' + \frac{q}{x^2}y = 0, \quad (6)$$

so that $P(x) = p/x$ and $Q(x) = q/x^2$ implying that the origin is a regular singular point whenever the constants p and q are not both zero.

Making the substitution $z = \ln x$, equation (5) gets transformed to

$$\frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + qy = 0, \quad (7)$$

which has the auxiliary equation

$$m^2 + (p-1)m + q = 0. \quad (8)$$

If the roots of (8) are denoted by m_1 and m_2 , then equation (7) has the following linearly independent solutions:

$$e^{m_1 z} \text{ and } e^{m_2 z} \quad m_1 \neq m_2,$$

$$e^{m_1 z} \text{ and } ze^{m_1 z} \quad m_1 = m_2.$$

Since $e^z = x$, then the corresponding solutions of equation (5) are

$$\left. \begin{array}{ll} x^{m_1} & \text{and } x^{m_2} \\ x^{m_1} & \text{and } x^{m_1} \ln x \end{array} \right\} \begin{array}{l} \text{if } m_1 \neq m_2, \\ \text{if } m_1 = m_2. \end{array} \quad (9)$$



The most general differential equation with a regular singular point at the origin is simply equation (6) with the constants p and q replaced by power series $p(x)$ and $q(x)$, respectively, as follows:

$$y'' + \left(\frac{p_0 + p_1x + p_2x^2 + \cdots}{x} \right) y' + \left(\frac{q_0 + q_1x + q_2x^2 + \cdots}{x^2} \right) y = 0. \quad (10)$$

If the transition from (6) to (10) can be accomplished by replacing constants by power series, then it can be guessed that the corresponding transition from (9) to the solutions of (10) can be accomplished by replacing power functions x^m by a series of the form (4), i.e., a Frobenius series.

Therefore, it is expected that (10) will have two linearly independent solutions of the form (4) or may be one of form (4) and one of the following form:

$$y = (a_0 + a_1x + a_2x^2 + \cdots)x^m \ln x, \quad (11)$$

where it is assumed that $x > 0$.

The indicial equation, in order to find the values of m , of the more general differential equation (10) is (obtained by equating the coefficient of x^{m+k-2} with $k = 0$ and $a_0 \neq 0$)

$$m(m-1) + mp_0 + q_0 = 0. \quad (12)$$

Why $a_0 \neq 0$ in series solutions?

If we put $a_0 = 0$ in expressions like $\sum_{k=0}^{\infty} a_k x^{m+k}$ or $(\sum_{k=0}^{\infty} a_k x^k) x^m \ln x$, then some positive integral power of x can be factored out of the power series part and combined with x^m .

Therefore, we assume that $a_0 \neq 0$ for such expressions. Its meaning is only that the highest possible power of x is otherwise factored out before any calculations are carried out.

Frobenius series evidently includes power series as special cases, whenever m is zero or a positive integer.

Example:

$$2x^2 y'' + x(2x + 1)y' - y = 0.$$

By using Frobenius series solution (since $x = 0$ is a regular singular point),

from indicial equation, $m_1 = 1$, $m_2 = -1/2$.

Now calculate $a_1, a_2, a_3, \dots$ in terms of a_0 .

For $m_1 = 1$, get $a_1 = -(2/5)a_0$, $a_2 = (4/35)a_0$ etc.

For $m_2 = -1/2$, get $a_1 = -a_0$, $a_2 = (1/2)a_0$ etc.

The solution can be obtained as

$$y = c_1 x \left(1 - \frac{2}{5}x + \frac{4}{35}x^2 + \dots \right) + c_2 x^{-1/2} \left(1 - x + \frac{1}{2}x^2 + \dots \right).$$

The gamma function $\Gamma(z)$ is defined by the improper integral

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt, \quad z > 0. \quad (13)$$

Note that

- The factor $e^{-t} \rightarrow 0$ so rapidly as $t \rightarrow \infty$ that this improper integral converges at the upper limit irrespective of the values of z .
- At the lower limit, $e^{-t} \rightarrow 1$, and the factor $t^{z-1} \rightarrow \infty$ whenever $z < 1$.
- The restriction that $z > 1$ is necessary so that convergence is guaranteed at the lower limit.

Basic Properties

- $\Gamma(z+1) = z \Gamma(z)$
- $\Gamma(1) = 1$
- $\Gamma(1/2) = \sqrt{\pi}$
- $\Gamma(n+1) = n!$ for any integer $n \geq 0$.